

Day 19 - Lecture 19.16: Best Practices for Data Visualization

1. Overview: The Synergy of Matplotlib & Seaborn

The single most recommended industry workflow combines **Matplotlib's layout control** with **Seaborn's statistical power**:

```
fig, ax = plt.subplots(figsize=(8, 5))
sns.lineplot(data=df, x='col1', y='col2', ax=ax) # Pass ax=ax
ax.set_title(...)                               # Fine-tune with ax methods
```

2. The Chart Selection Decision Matrix



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3. Core Rules for Professional Visualizations

1. Maximize the Data-to-Ink Ratio (Edward Tufte):

- Eliminate unnecessary borders, 3D effects, and garish background colors.
- Use subtle gridlines (`linestyle=':'` , `alpha=0.5`).

2. Accessible Color Choices:

- Avoid pure red-green combinations without shape/marker differences (affects 8% of male population with deuteranopia).
- Use perceptual palettes: `colorblind` , `viridis` , or `cividis` .

3. Always Label Units:

- Instead of `plt.xlabel("Revenue")`, use `plt.xlabel("Revenue (in $ Millions)")`.

4. Sort Categorical Data:

- Unordered bars force the eye to jump back and forth. Always sort categories by value unless there is a natural temporal order (e.g. Mon, Tue, Wed).

4. Code Implementation: The Recommended Pattern

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

# Create figure and axes explicitly using OO style
fig, ax = plt.subplots(figsize=(9, 5))

# Plot high-level Seaborn chart onto target axes
sns.lineplot(
    data=tips,
    x="day",
    y="total_bill",
    hue="sex",
    marker="o",
    markersize=8,
    errorbar=None,
    palette=["#2b5c8f", "#d95f02"],
    ax=ax
)

# Customize precisely using Axes methods
ax.set_title("Average Total Bill by Day and Sex", fontsize=14, fontweight="bold",
ax.set_xlabel("Day of Week", fontsize=12)
ax.set_ylabel("Total Bill ($)", fontsize=12)
ax.set_ylim(0, 26)

# Refine legend
ax.legend(title="Customer Sex", frameon=True, facecolor='white', framealpha=0.9)
ax.grid(axis='y', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
```

5. Complete Solutions to Day-19 Assignment Problems

Problem 1: Seaborn Tips Box Plot by Day & Sex

- **Task:** Create box plot of `total_bill` for each `day`, add `hue="sex"`, identify day with highest median bill, comment on outliers.
- **Answer:** Sunday (`Sun`) exhibits the highest median total bill (~ 20). *Multiple high outliers (>40)* are observed on Saturday and Sunday nights.

Problem 2: Total Bill vs Tip Scatter Plot

- **Task:** Color points by `sex`, marker styles by `smoker`, analyze if smokers tip differently.
- **Answer:** Smokers display greater variance in tipping behavior (some tip exceptionally high, others tip low), whereas non-smokers show a tighter linear relationship with bill size.

Problem 3: Normal Distribution & Passing Threshold

- **Task:** Generate `np.random.normal(70, 10, 200)`, plot histogram, draw `axvline` at mean.
- **Answer:** Visualizes bell-shaped symmetric Gaussian distribution where Mean = Median = Mode ≈ 70 .

Problem 4: Average Tip Heatmap Pivot Table

- **Task:** Pivot table of average tip with Rows= `day`, Columns= `time`, annotated heatmap.
- **Answer:** Sunday Dinner delivers the highest average tip.